# Market Basket Analysis with Python

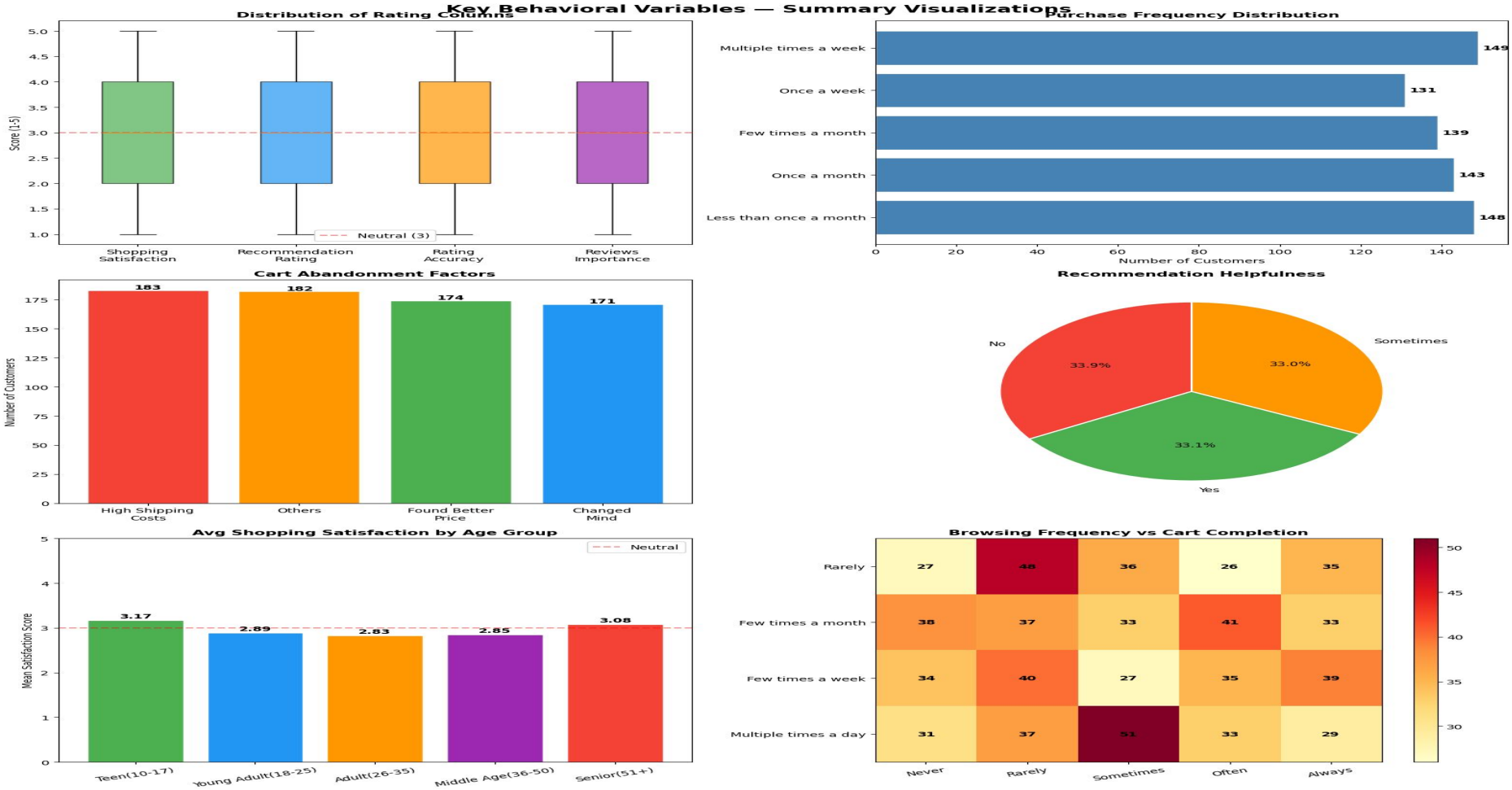
# Task 1: Data Cleaning and Preparation

- Upon cleaning, we found there are total of final 710 rows and 24 columns.
- The time related data have been standardized and have been converted to datetime, extracting Date, Hour, DayOfWeek.
- Dataset has now got cleaned with 710 rows and 33 columns with zero null values.
- The column with name Personalized_Recommendation_Frequency have been transformed to Personalized_Recommendation_Rating,
- All four rating columns have a mean around 3.0 and are perfectly distributed across the 1 - 5 scale with no nulls.

# Task 2: Descriptive Behavior Analysis

- The mean age is 38.4 years and the median age is 37 years with age range 10 to 67 years.
- All categories are nearly equally popular with 49-53% customers purchasing from each indicating customers are multi-category shoppers rather being loyal to one category.
- Browsing frequency is almost perfectly evenly spread across all four levels that is suggesting no dominant browsing pattern exists among Target customers.
- Teens are most affected by high shipping costs relative to their group size.
- All four metrics hover around 3.0 suggesting that customers are neither highly satisfied nor dissatisfied.

- Teens are most satisfied, likely due to lower expectations.



# Task 3: Customer Segmentation and Profiling

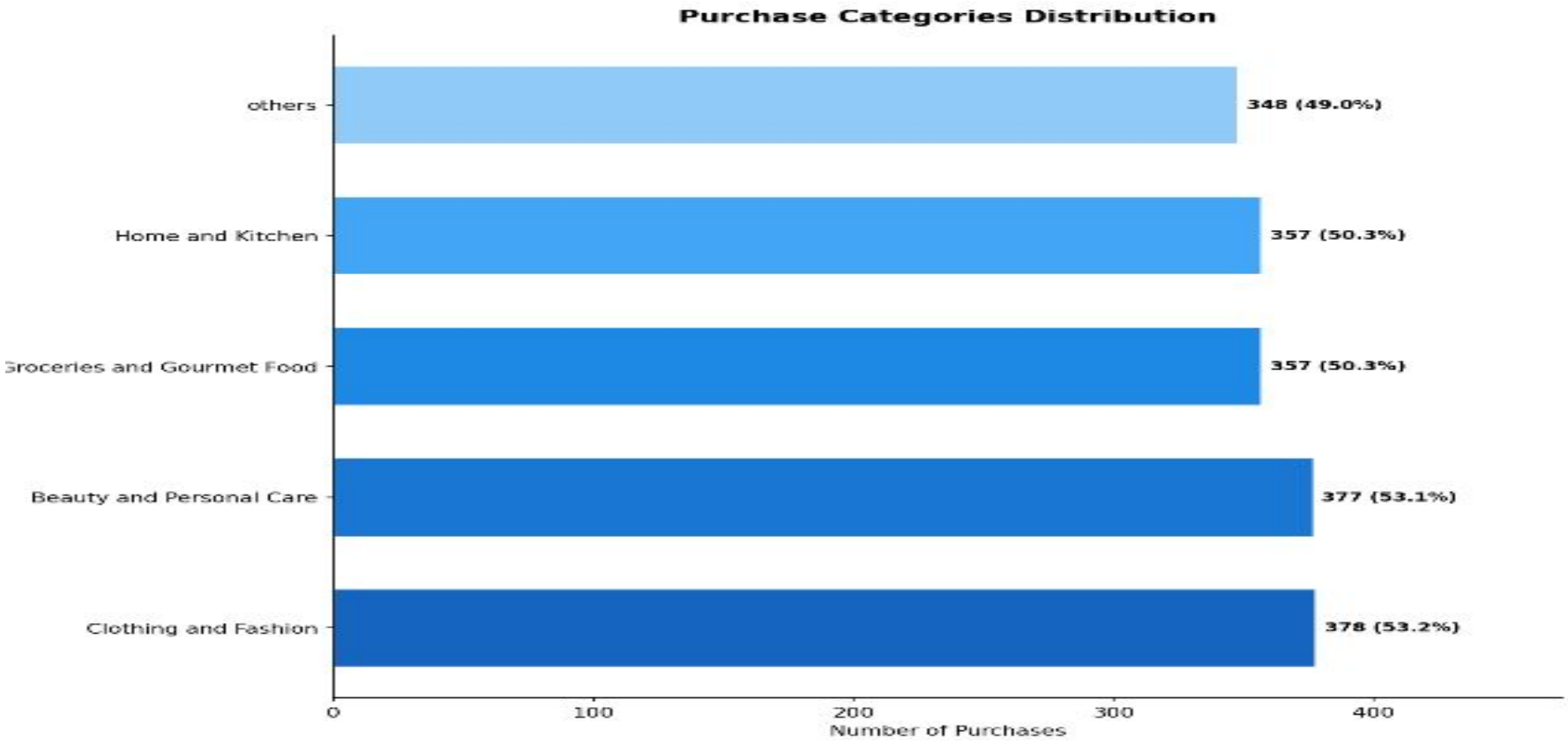
- Customer base is polarized, out of which majority are either low or high frequency shoppers with medium frequency being the smallest group.
- Count of low satisfaction is 293, followed by 152 of medium satisfaction and 265 of high satisfaction.
- High frequency customers have above average rating accuracy perception.
- The average satisfaction of at-risk customers is 2.37, the average satisfaction of frequent buyers is 4.54, the average satisfaction of potential converters is 35.18
- Occassional shoppers find recommendations most helpful.
- Frequent buyers find recommendations least helpful.
- At-risk customers perfectly split recommendations that is neither help nor hurt retention.

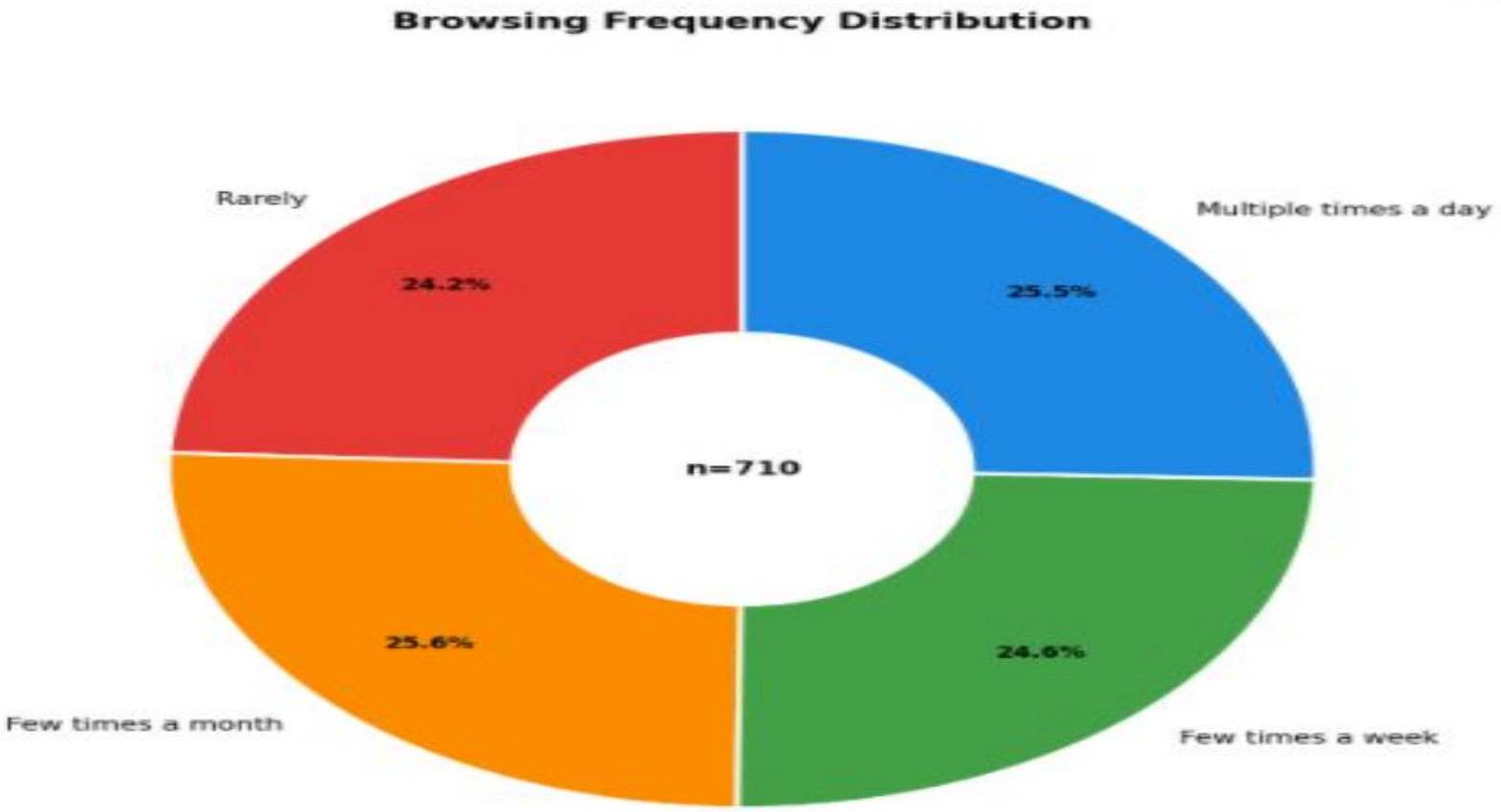
# Task 4: Recommendation and Review Insights

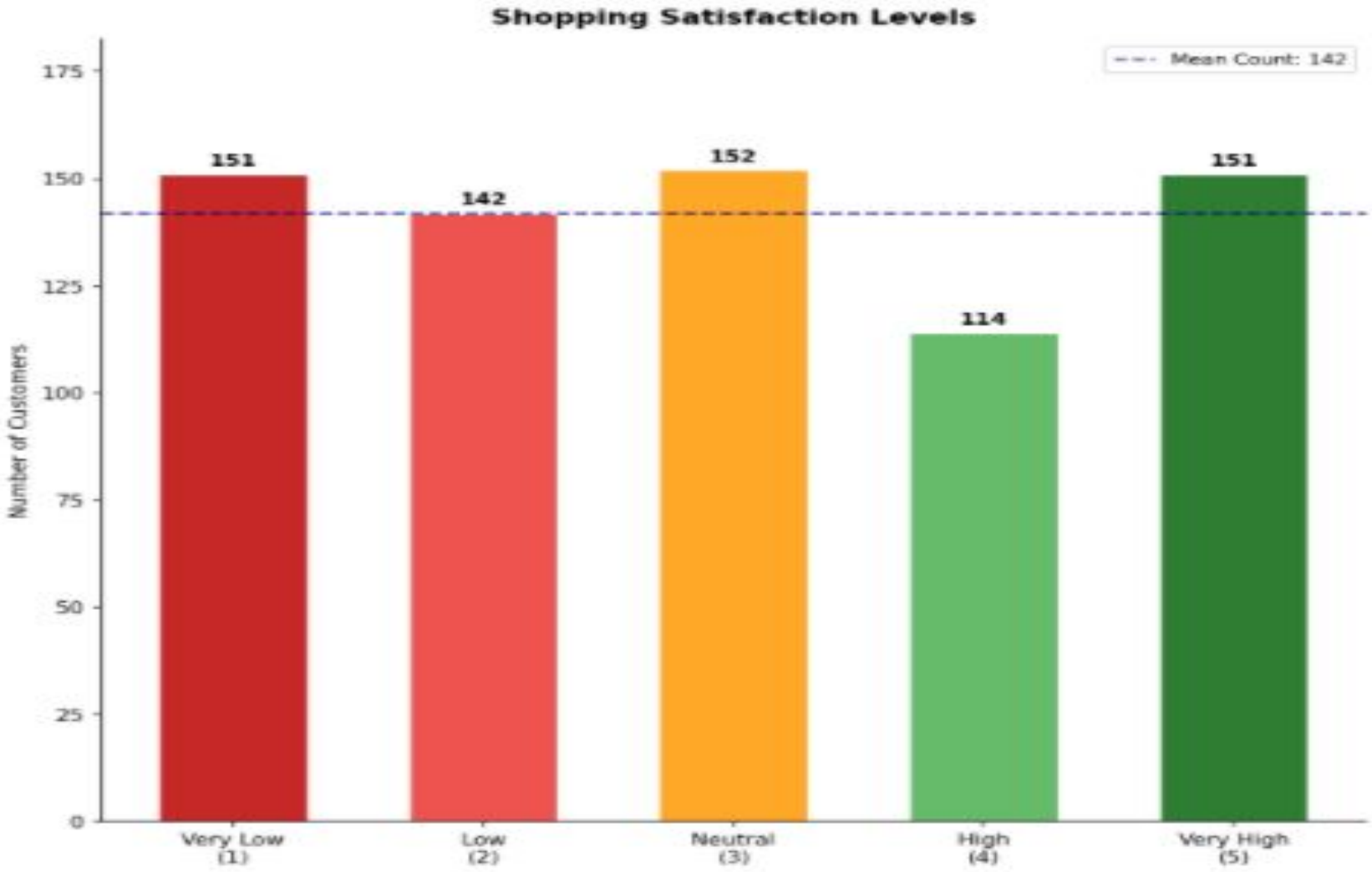
- “No” helpfulness group has the most Score 1 responses, “Sometimes” group has most Score 5 responses.
- Customers who always receive recommendations are least satisfied.
- Occasional review reliers are most satisfied as they have balanced trust without over dependence.
- Moderate review reliers are least satisfied as the mid-level trust yields worst outcomes.
- Customers who find reviews most helpful have highest expectations and are harder to satisfy.
- Customers who did not leave reviews are slightly more satisfied in comparison to those who did.

- All age groups show nearly identical distributions across No/Sometimes/Yes.
- Adults engage most with recommendations.
- Teens and Seniors are most likely to reject recommendations.
- Recommendation frequency stays flat across all purchase frequencies.
- Customers who engage with recommendations complete more carts but are least satisfied.
- Customers who find recs “Yes” helpful have lowest purchase frequency and the shoppers need recommendations the most.
- Any recommendation improvement must be paired with pricing strategy reforms to show meaningful satisfaction gains.

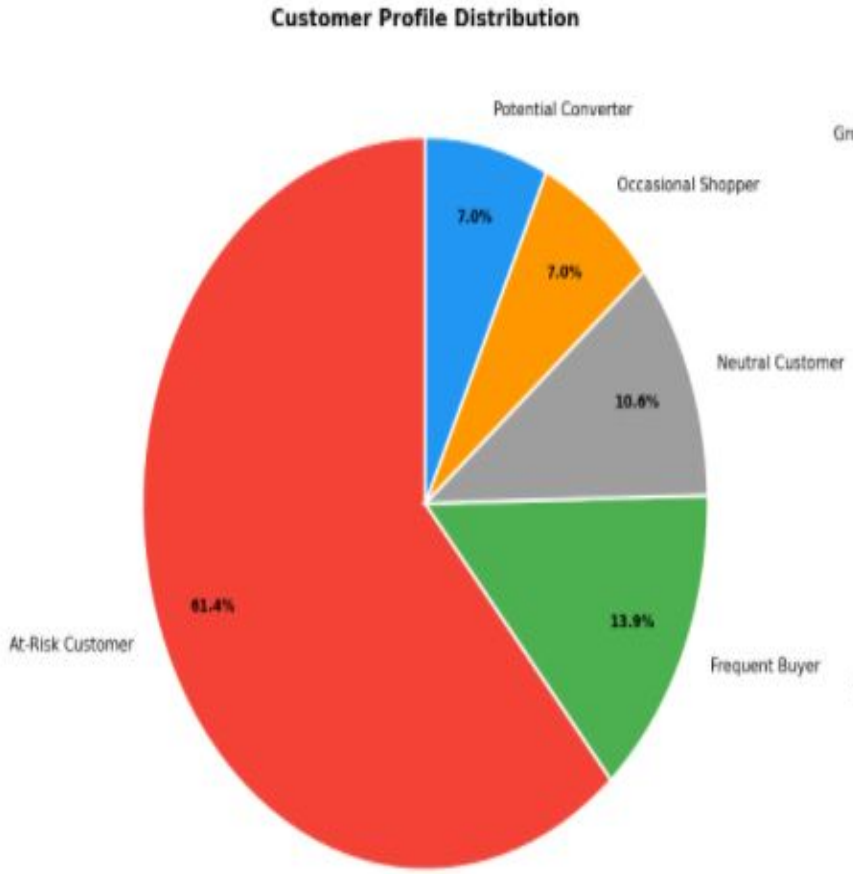
# Task 5: Visualization and Reporting

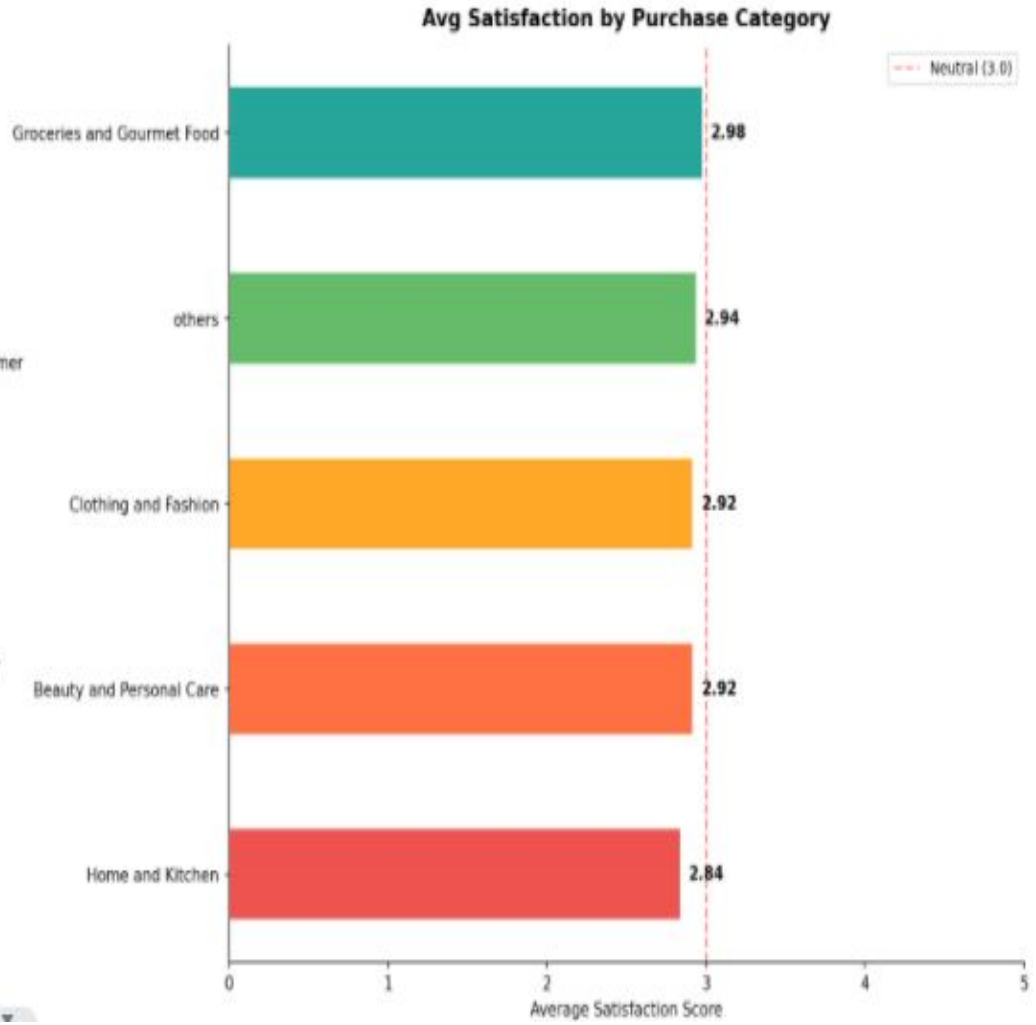



Browsing Frequency Distribution
Rarely
24.2%
Multiple times a day
25.5%
n=710
25.6%
Few times a month
24.6%
Few times a week


Shopping Satisfaction Levels
Mean Count: 142
Number of Customers
175
150
125
100
75
50
25
0
151
142
152
114
151
Very Low
(1)
Low
(2)
Neutral
(3)
High
(4)
Very High
(5)

Recommendation Helpfulness vs Satisfaction
Recommendation Helpfulness
Yes
Sometimes
No
47
50
54
40
44
48
39
55
38
54
56
53
43
36
53
Very Low (1)
Low (2)
Neutral (3)
High (4)
Very High (5)
Satisfaction Score
55.0
52.5
50.0
47.5
45.0
42.5
40.0
37.5


Customer Profile Distribution
Potential Converter
7.0%
Occasional Shopper
7.0%
Neutral Customer
10.6%
Frequent Buyer
13.9%
At-Risk Customer
61.4%


Avg Satisfaction by Purchase Category
Neutral (3.0)
Groceries and Gourmet Food
2.98
others
2.94
Clothing and Fashion
2.92
Beauty and Personal Care
2.92
Home and Kitchen
2.84
0
1
2
3
4
5
Average Satisfaction Score

# Task 6: Video Presentation (20 Marks)

The link for the video has been pasted below:

https://www.loom.com/share/f14e074de70d472ea5efebfb848fd2ad